

Code-Layout, Readability and Reusability

Date	15 July 2026
Team ID	SWUID2026217449
Project Name	Rising Waters - Flood Prediction ML
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout	Yes	4-space indentation used consistently
2	Proper File Structure	Files and folders are logically organize	Yes	Organized into app.py, templates/, and model files
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables like annual_rainfall, cloud_visibility used
4	Function / Method Names	Functions are descriptively named	Yes	Functions like predict() are clearly named
5	Code Comments	Inline and block comments explain log	Yes	Comments added for each processing step
6	Modular Design	Code is split into reusable functions/m	Yes	Data preprocessing and model logic kept modular
7	No Redundant Code	Duplicate or unused code is removed	Yes	No duplicate code blocks found
8	Error Handling	Exceptions and errors are handled grand	Partial	Basic try/except handling can be extended further

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Function	Python	Used in both training and prediction	High

2	Model Loading Utility	Python	Used in app.py for predictions	High
3	Scaler Transform Function	Python	Used before every prediction	High
4	HTML Form Template	HTML	Reused across index and result pages	Medium
5	Flask Route Handler	Python	Reused for form submission logic	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clean, organized folder and file structure
Readability	5	Clear variable and function names throughout
Reusability	4	Core functions reused across training and app
Documentation / Comments	4	Well-commented notebook cells
Overall Score	4.5	Strong overall code quality